Weakly Supervised Road Segmentation in High-Resolution Remote Sensing Images Using Point Annotations

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***Abstract*— Road segmentation methods based on deep neural networks have achieved great success in recent years, but creating accurate pixel-wise training labels is still a boring and expensive task, especially for large-scale high-resolution remote sensing images (HRSIs). Inspired by the stacked hourglass model for human joints detection, we propose a weakly supervised road segmentation method using point annotations in this article. First, we design a patch-based deep convolutional neural network (DCNN) model for road seeds and background points detection and train the model using point annotations. Then, in the process of road segmentation, the DCNN model detects a series of road and background points that are used to train a Support Vector Machine Classifier (SVC) for classifying each pixel into road or nonroad. According to the local geometry of road and the inaccurate classification of SVC, a multiscale and multidirection Gabor filter (MMGF) is put forward to estimate the road potential. Finally, the active contour model based on local binary fitting energy (LBF-Snake) is introduced to extract the road regions from the inhomogeneous road potential. Qualitative and quantitative comparisons show that our method achieves results close to the fully supervised semantic methods without considering the annotation cost and outperforms them given a fixed budget.**

***Index Terms*— Active contour model, Gabor filter, point annotation, remote sensing images, road segmentation, weakly supervised segmentation.**

I. Introduction

ROAD segmentation in the remote sensing images is the hot research issue due to its vital applications in cartography, traffic planning, vehicle navigation, intelligent transportation system, and so on. Despite a huge number of works dealing with this problem, the results are still limited and not generic enough to solve the lack of information due to the existence of artifacts such as overcast of building shadows, roadside trees, or cars [1]. With the development of computer vision, deep learning technologies have been widely used in the field of road segmentation. However, training these deep learning models demands massive training samples, while annotating such a large number of samples is a tough, boring, and expensive task. In order to reduce the label cost of training data, weakly supervised semantic segmentation has become a research hotspot in recent years.

Currently, weakly supervised road segmentation in remote sensing images mainly refers to the study of road segmentation model trained by imprecise data. The imprecise data that are generated from existing maps (such as OpenStreetMap) [2], [3], [4] usually suffered from missing and registration noise. Mnih and Hinton [2] designed a patch-based deep convolutional neural network (DCNN) model to label the roads in aerial images, and the model was trained by the inaccurate data generated from OpenStreetMap. The road references are often incomplete and poorly registered; thus, Mnih [3] further proposed two loss functions to deal with the label noise. Similarly, Wu *et al.* [5] extracted road regions only using the road centerline generated from OpenStreetMap, First, the road annotations were inferred according to the prior knowledge of road width and the image resolution. Then, the road segmentation was regarded as a weakly supervised semantic segmentation using scribble labels by optimizing a regularized semisupervised loss. However, for the places where the map is unavailable, the road ground truth still needs to be labeled by the time-consuming and laborious manual operation.

Generally, when we discuss weakly supervised semantic segmentation, we refer the weak labels such as image-level class labels, points, bound boxes, and scribbles [6]. In road segmentation field, the image-level class labels are infeasible because there is only one object class (i.e., road) except for background, and the bounding boxes are also useless because of the wide spatial extent of road network. On the other hand, scribbles seem to be suitable labels for road annotation. However, drawing graffiti on an elongated road is somewhat troublesome, because it is prone to draw graffiti outside the road area using computer mouse. Finally, point annotations are applied for road segmentation in our method. However, according to the sparse distribution of road, marking a road object only by a point (as done in [7], where an object was labeled by a point) is impossible to extract the whole road

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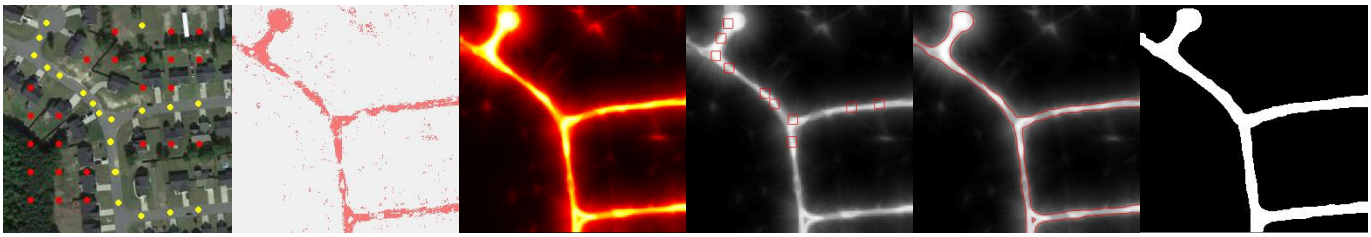


Fig. 1. (a) Test image with road seeds (yellow) and background seeds (red). (b) Pixel classification achieved by SVC trained by these seeds, in which red and white represent road and nonroad classes, respectively. (c) Road potential estimated by Multiscale and Multidirection Gabor kernels. (d) Initial contours generated from seeds. (e) Road contours achieved by LBF-Snake according to road potential. (f) Road segmentation by simple post-processing.

object out of the complex context, especially from the large high-resolution remote sensing image (HRSI). Thus, we label a road segment with multiple points, and we call it the enhanced point-supervision.

Inspired by the human joint points detection in human pose estimation and the local geometric features of road in HRSI, a weakly supervised road segmentation framework based on point-level labels is proposed in this article. This framework combines DCNN and heuristic algorithms to obtain the final road regions according to the following process. First, a patch-based Road Seeds Estimation Model (RSEM) is used to search the road seeds and background points in the test image, which is trained in advance by point annotations sampled from training images. Second, these road and background points are regarded as positive and negative samples to train a Support Vector Classifier (SVC) which in turn classifies each pixel of the image into road or nonroad. Third, the road probability density map (road potential) is estimated by a multiscale and multidirection Gabor filter (MMGF) based on the imperfect segmentation inferred by SVC. Fourth, an active contour model algorithm is introduced to extract the road contours, and finally, the road regions are extracted and refined by a simple post-processing. The pipeline is illustrated in Fig. 1 and detailed in Section III. Note that, RSEM is trained by point annotations in advance that is why we named our approach as weakly supervised road segmentation. RSEM automatically detects a large number of road and background points for the training of SVC, which overcomes the difficulty of manual sampling. Our approach combines the DCNN model and heuristic algorithms to detect roads in remote sensing images. The powerful feature representation of DCNN and the interpretability of heuristic algorithm ensured the effectiveness of our method. To the best of our knowledge, this work is the first attempt to introduce point-supervision into road segmentation in high-resolution remote sensing images.

II. Related Work

The goal of weakly supervised semantic segmentation is to leverage weak annotations instead of pixel-wise ones to train deep learning models for semantic segmentation with minimum human supervision. According to the accuracy, from weak to strong, the weak labels can be classified into image-level class labels, points, bound boxes, and scribbles [6]. These labels are easier to be collected than pixel-wise labels as they require much fewer annotation costs. Image-level class labels indicate the presence or absence of entities in an image missing the locations and the scopes of them. Point label is an instance-wise point, which roughly points out the center location of an object, but does not indicate the scope of the object. The bounding box offers the location and the scope of an object by a rectangular box tightly covering the entire object. A scribble roughly provides the location and the extension of an object with a single stroke. The image-level class label has been widely used because it demands the least manual annotation costs and is already available in existing large-scale data sets (such as ImageNet [8]). The key problem of image-level weakly supervised semantic segmentation is to find the association between class labels and pixels. In [9], an image-level weakly supervised semantic segmentation method named SEC was proposed, which formulated the methodology of weakly supervised semantic segmentation, namely three basic principles: Seed, Expand, and Constrain. That is, finding the location (i.e., seed) of an object using Class Activation Mapping (CAM) [7]–[8], and then expanding the seed area while constraining the expansion degree to make the segmentation consistent with the object boundary. Point label has strengthened supervision than image-level label because it points out where the object is located. Bearman *et al.* [7] proposed a loss function to train a convolutional neural network (CNN) model exploiting point supervision and objectness potential. Bounding box labels are relative strong supervision annotation with still low labeling costs. Papandreou *et al.* [10] trained a model for semantic segmentation using the fake training data which was estimated by GrabCut [11] according to the box supervision. For scribble annotation, Di *et al.* [12] formulated the label estimation as an optimization problem, and the model training was conducted by the alternation between label estimation and the learning of the model parameters.

Currently, most of the researches focus on the weakly segmentation in natural images, such as the works on PASCAL VOC [13], Cityscapes [14], COCO [15], and MSRC 21 [16]. However, in the research on road segmentation in HRSI, most researchers study road segmentation as fully supervised semantic segmentation, regard the noisy labels as accurate ones to train fully supervised models, and achieve good results. The explanation is that the road references generated from map have a certain degree of accuracy, and the DCNNs can grasp the color and geometric features of roads from such noisy data. Zhong *et al.* [17] introduced the Fully Convolutional Network (FCN) [18] into road segmentation,

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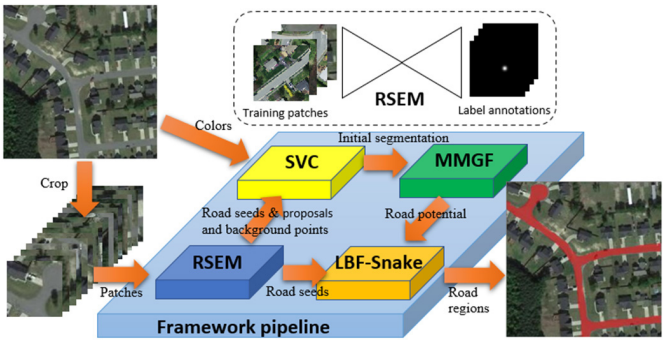


Fig. 2. Overall framework of the proposed approach.

and discovered that the result could be significantly improved by adding the outputs of pooling layers into the final score layer. More methods were proposed based on encoder–decoder architecture to alleviate the by-effect of the interpolation upsample operation in FCN. Panboonyuen *et al.* [19] presented an enhanced deep convolutional encoder–decoder network for road segmentation in aerial images based on SegNet. Mattyus *et al.* [20] developed a variant of FCN using ResNet [21] as encoder and deconvolutional layers as decoder to estimate road topology from aerial images directly. Zhang *et al.* [22] designed a DCNN combining the strengths of residual learning and U-Net to extract road areas. In some works [23], [24], [25], [26], weakly supervised data (i.e., road centerline) and precise road mask were simultaneously exploited to enhance the road segmentation, and the experiments showed that the road structure and connectivity indicated by road centerlines could boost road segmentation.

Some researchers study semisupervised road extraction, that is, accurately label some road areas in an image, then estimate other road areas using optimization algorithms, because, in most case, we only have a small set of labeled data and a large collection of unlabeled ones. Cheng *et al.* [27] proposed a semisupervised method to obtain the road regions by exploring the intrinsic structures between the labeled samples and the unlabeled ones.

III. Methodology

The overall pipeline of the proposed approach is presented in Fig. 2. First, a series of patches are cropped from the test image by a sliding window with a certain step, which will be fed into RSEM (introduced in Sections III-A and III-B). The RSEM outputs a number of road and nonroad seeds companied with road proposals [see Fig. 10(i)–(l)]. Second, seeds and the test image are used to train an SVM classifier (SVC; see Section III-C), which in turn classifies each pixel of test image into road and nonroad, called preliminary segmentation. The road proposals predicted by RSEM are also added as road pixels into preliminary segmentation to reduce the false negatives (FN) in the complex scenarios. Third, the multiscale and multidirection Gabor filter (MMGF; described in Section III-D) estimates road potential according to the preliminary segmentation. Fourth, taking road potential as input and the rectangles centered at road seeds as initial contours, local binary fitting (LBF)-Snake (detailed in Section III-E) deforms the contours to the road boundaries. Finally, the road regions are refined by a simple post-processing.

A. Road Seeds Estimation Model

In this article, we try to detect road regions using point labels. The road network presents a narrow and elongate structure and generally extends to the entire image. For such a sparse distribution of roads, the existing point-supervised deep learning networks are difficult to recognize the entire road regions according to a point, so we propose a patch-based RSEM which is used to predict road center points for a serial of patches in our method. Thus, we can get a number of road center points throughout the image. On the other hand, we can also obtain a lot of background points according to the output of the model. These road centers and background points are regarded as seeds (road and nonroad) for the subsequent heuristic algorithms for road recognition. The road seed estimation model has been presented and discussed in our previous research [28]. In order to facilitate reading and understanding, we briefly introduce the network structure and principles of the model in this section. Inspired by the study of human pose estimation, we regard the road center points as special points of a small patch. The classic model for key points estimation is the Stacked Hourglass Network (SHN) proposed by Newell *et al.* [29], which is an FCN used to predict the joints of human body. The key component in the network is the hourglass model, a cascaded multiscale top-down to bottom-up model for context feature representation. There are several hourglass models stacked in the network to achieve better feature extraction. The network outputs a confidence map indicating the confidence of each pixel as a human joint. We redesign the SHN network according to our task. The detail configuration is shown in Fig. 3. Meanwhile, we propose a weighted cross entropy loss (WCELoss) function to deal with the extremely label unbalance instead of using the original mean square error loss (MSELoss), because just a very small number of pixels around the road centers are regarded as positive samples. The WCELoss is shown in (1), where N is the pixel number in the patch, Y is the ground truth which is a Mixed Gaussian map centered at road centers, and n is the number of road segment inside the patch, while (x_c, y_c) is the cth road center, i.e., a road point is sampled from a road segment. y_i is the ith pixel's ground truth, and $\hat{y}_i$ is the prediction of the ith pixel. Thus, e^{y_i} acts as an adaptive weight to penalize the predicting errors near the road centers. λ is a scale factor used to determine the ratio of the loss coming from the errors near the road centers. Higher λ makes RSEM output higher global confidence. Since the road center points are selected by the local peaks, it has nothing to do with the global confidence, and we set $\lambda = 1$ in all experiments. Table I shows the improvements of WCELoss. The road center confidence predicted by RSEM that trained by WCELoss has larger spacing between road patches and nonroad patches, which is an important basis for threshold setting to distinguish the road patches from background ones during the road and nonroad points search. The spacing is calculated by (2). Larger spacing means more robustness

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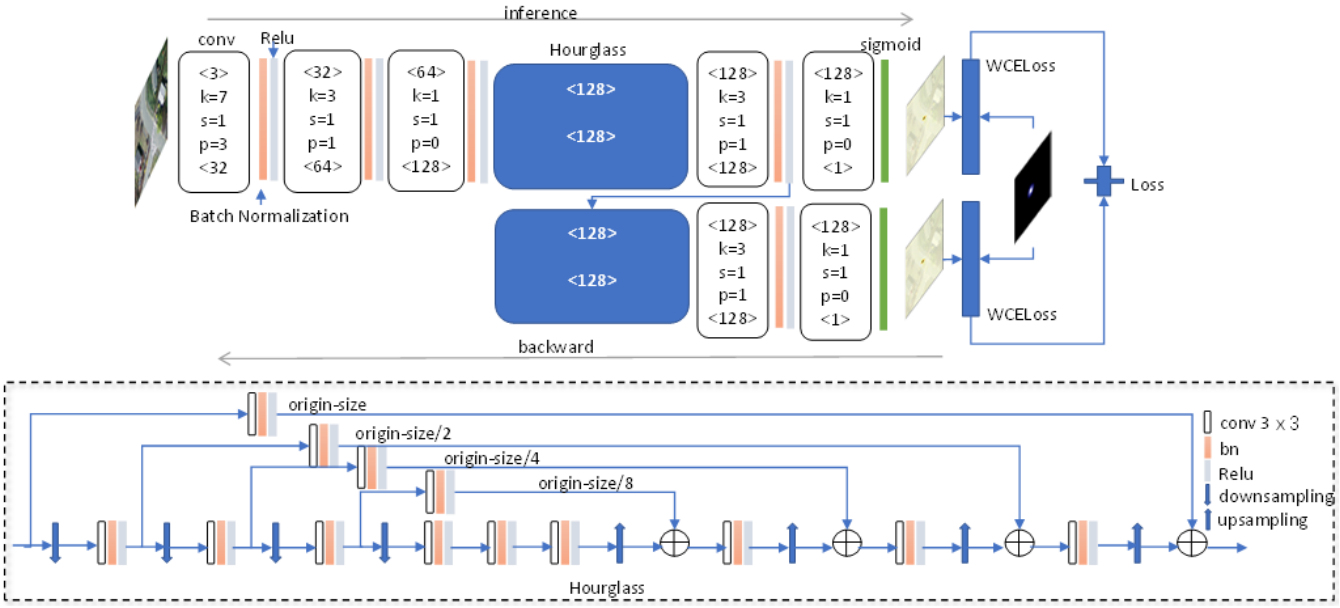


Fig. 3. Network architecture of the road center estimation model, where k, s, p, and $< n >$ are the kernel size, stride, padding, and channels, respectively.

TABLE I

COMPARISONS BETWEEN WCELOSS AND MSELOSS, WHERE μ_W AND μ_{Wo} ARE THE MEAN VALUES OF THE MAX CONFIDENCES OF PATCHES WITH ROAD AND WITHOUT ROAD, AND THE SAME TO σ_W AND σ_{Wo}, WHERE σ MEANS STANDARD DEVIATION. THE F1-SCORES ARE THE PERFORMANCES ACHIEVED BY RSEM TRAINED BY DIFFERENT LOSS FUNCTIONS, WHICH IS COMPUTED AS MENTIONED IN SECTION IV-F. THE COMPARISON IS CONDUCTED ON THE GOOGLE EARTH DATA SET

Loss type	μ_w	σ_w	μ_{wo}	σ_{wo}	Spacing	F1-score
MSELoss	0.2397	0.0698	0.0838	0.0620	0.0241	0.9036
WCELoss	0.4284	0.0974	0.0885	0.1041	0.1384	0.9514

of the threshold setting. Meanwhile, the RSEM trained by WCELoss performs better than that trained by MSELoss (see F1-score in Table I)

$$\text{WCELoss} = \frac{1}{N}\sum_{i=1}^{N} e^{y_i}[\lambda y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)]$$

$$\text{where} \quad y_i \in Y = \sum_{c=1}^{n} e^{-\frac{[(x-x_c)^2+(y-y_c)^2]}{\sigma^2}} \tag{1}$$

$$\text{spacing} = (\mu_w - \sigma_w) - (\mu_{wo} + \sigma_{wo}). \tag{2}$$

B. Road Seeds and Background Points Search

A sliding window is used to collect patches with the size of $N \times N$ at a fixed step M in terms of top to bottom and left to right. $N \times N$ is the size of the training patches that used to train the road seed estimation model previously. The value of M can be adjusted according to tasks. The smaller M, the more patches are collected, and the more seeds are obtained, but higher computation is required. Generally, M can be between $N/4$ and N. In this article, M is set to $N/2$. The RSEM outputs a road center confidence map by feeding an image patch. Then, the candidate road center points are obtained using the nonmaximum suppression (NMS) algorithm. We set dual thresholds to filter these candidate points, that is, the candidate points are taken as a road center points when their confidence values exceed the high threshold and there is no road in the patch if all the confidences of these candidate points are below the low threshold, and the center point of the patch is taken as a background point. The effect is shown in Fig. 1(a), where the red dots are the background points and the yellow dots are the road seeds.

C. Pixel Classification

For each test image, a Support Vector Machine Classifier (SVC) is trained by taking road seeds and background points as positive and negative samples. SVC is a classification technique developed by Vapnik and his group [30], [31]. The main idea of SVC is to separate data by a hyperplane by maximizing the margin among them [32]. The decision boundary should be nonlinear due to the mixed spectral signatures of each pixel. We map the feature vectors to a higher dimensional Euclidean space by a mapping function Φ. Meanwhile, in order to tolerate some outlier samples, the slack variables ξ_i are introduced into the objective function to construct a soft margin, as shown in (3), where w is an n-dimensional vector perpendicular to the hyperplane, b is the distance of the closest point to its project point on the hyperplane, and y_i is the class label for ith sample which equals 1 for positive samples and -1 for negative ones.

Using the trained SVC, the class of each pixel can be predicted by (4) according to its color feature, where λ_i is the Lagrange Multiplier introduced in the solution of the SVC problem. Only λ_i corresponding to the support vectors are greater than 0 and all others are 0. We use Radial Basis Function (RBF) (5) as kernel function for all the images in both test data sets, because we found it performs well in various scenarios (see Section IV-D for details). It is worth noting that we use the Luv color features, because the pixels are easier discriminated in Luv color space. Fig. 1(b) shows the

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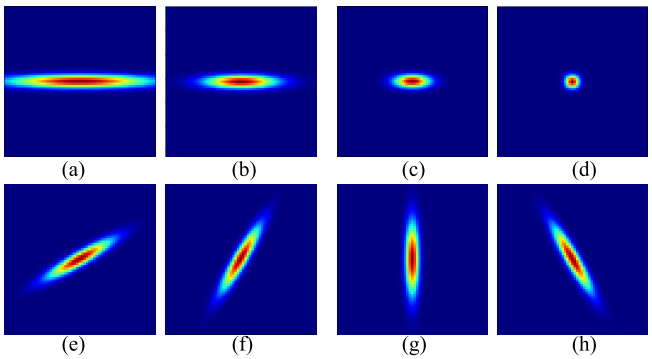


Fig. 4. Exhibition of some kernels in the MMGF. (a)–(d) show the multiscale of the 0-C kernels, and the γ parameters are 0.1, 0.2, 0.4, and 1.0, respectively. (e)–(h) display the some of the 0.2-γ kernels, and the directions are 30°, 60°, 90°, and 120°, respectively.

pixel classification of Fig. 1(a), where red and white represent road and nonroad. We emphasize that unlike segmentation methods based on superpixels, we classify pixels instead of superpixels [33], [34], because the classification results based on superpixel seriously rely on the superpixel segmentation. However, the superpixels generally cannot nicely fit the road-sides in the HRSI due to the low contrast between the road and the background in most cases, and worse to the regions covered by interferences

$$\min_{w,b,\xi}\left(\frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}\xi_i\right) \tag{3}$$

$$\text{s.t. } 1 - y_i(w^T\Phi(x_i) + b) - \xi_i \le 0, \xi_i \ge 0$$

$$f(x) = \text{sign}\left(\sum_{i=1}^{n} y_i\lambda_i K(x, x_i) + b\right) \tag{4}$$

$$K(x, x_i) = \exp\left(-\alpha\|x - x_i\|^2\right). \tag{5}$$

D. Road Potential Estimation

Due to the variance of road color and the occlusion of interferences, the road areas cannot be segmented nicely based on the color features. As shown in Fig. 1(b), it can be seen that some locations on the road surface are erroneously classified due to the color inhomogeneity, which is more serious in some places covered by trees, cars, shadows, and so on. According to the local geometry of the road, a road segment presents a straight stripe with a strong directional characteristic within a sufficient local range. When some pixels are wrongly classified, their collinear neighbors along the road direction can prove them as road members. However, the road directions are various and the gapes of the broken roads also vary. So, we propose MMGF to assess the road probability for each pixel according to the preliminary classification by SVC. The 2-dimensional Gabor filter is defined by (6), where (x, y) is a coordinate in the filter and λ, θ, ψ, σ, and γ denote the wavelength, direction, phase shift, Gaussian standard deviation, and the spatial aspect ratio, respectively. N is a normalization factor. In this article, we construct MMGF as a Gabor kernel bank in which θ ranges from 0° to 180° with interval 5, and γ ranges from 0.1 to 1 (see the experiment setting in Section IV for details). The Gabor kernels are narrower with smaller γ, which means that farther neighbors in the direction θ will

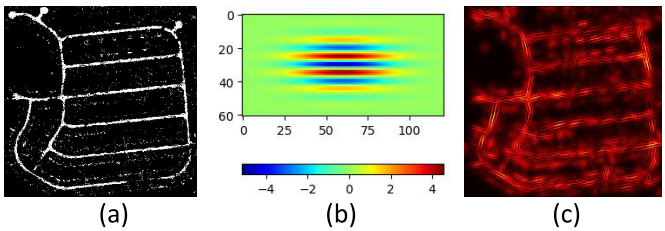


Fig. 5. (a) Pixel classification. (b) One kernel in the MMGF with striations. (c) Undesirable ripple effects in road potential filtered by rippled Gabor kernels.

be involved to verify the road membership of a pixel. Fig. 4 shows some kernel examples. On the other hand, the response of Gabor filter presents stripes around the road areas if the Gabor filter appears multistripes, as shown in Fig. 5, which is undesirable. So, we set σ to a small value and λ to a large value to prevent the appearance of striations. In our experiments, σ is set to 2 and λ is fixed to 30. The road probability estimator is defined as (7), where $f(i, j)$ is the class label of pixel (i, j), 1 for road and 0 for nonroad, and w and h are the width and height of a Gabor filter, respectively. It is worth noting that the convolution in (7) can be computed efficiently in Fourier spectrum, although the Gabor filters are very large. Fig. 1(c) shows the road probability density achieved by MMGF

$$g(x, y; \lambda, \theta, \psi, \sigma, \gamma) = \frac{1}{N}e^{-\frac{x'^2+\gamma^2y'^2}{2\sigma^2}}\cos\left(2\pi\frac{x'}{\lambda} + \psi\right) \tag{6}$$

$$\text{where } \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$$

$$\hat{f}(x, y) = \max_{\forall\theta,\forall\gamma}\left[\sum_{i=-w/2}^{w/2}\sum_{j=-h/2}^{h/2} f(x+i, y+j) \times g(i, j; \theta, \gamma)\right]$$

$$\text{s.t. } \theta \in [0, \pi), \quad \gamma \in [0.1, 1.0]. \tag{7}$$

E. Road Segmentation Based on LBF-Snake

Studying the road probability map, we find that in the areas with dense road pixels in the preliminary segmentation, the road potential is high, and vice versa, especially where a large number of pixels are wrongly classified. Fig. 6 shows an example of hard binarization on a road probability map, where the left-slanting road fails to be extracted completely and the two right-slanting parallel roads are wrongly segmented into one broad road. Thus, simple thresholding on the road probability density map fails to obtain a satisfactory road segmentation. Inspired by [35], we extract the road regions using LBF-Snake algorithm which can overcome the problems caused by intensity inhomogeneities. Given an image I, the local binary fitting energy for a given pixel x is defined as (8), where C is a closed contour which separates I into two regions: $\Omega_1 = \text{outside}(C)$ and $\Omega_2 = \text{inside}(C)$. $K_\sigma$is a Gaussian kernel function with the standard deviation σ, and $f_i(x)$ is the value that approximates the intensity of pixel x inside region Ω_i. The Gaussian kernel $K_\sigma(x - y)$ rapidly approaching 0 when $|y - x| > 3\sigma$, thus the fitting energy

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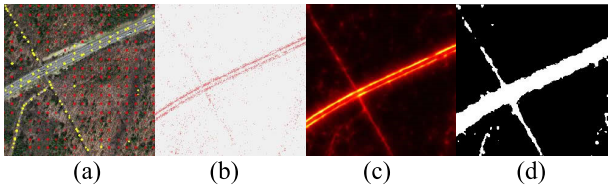


Fig. 6. Example shows that the variation of road potential in one image. (a) Image marked with seeds autosearched by road center estimation model. (b) Pixel classification by SVC. (c) Road potential estimated by MMGF. (d) Binarization of road potential by a simple threshold.

$\varepsilon_x^{\text{Fit}}$is localized around the pixel x. Given a pixel x, $\varepsilon_x^{\text{Fit}}$ can be minimized when C exactly on the object boundary and f_i optimally approximate the local intensity inside Ω_i. To obtain the entire object boundary, the global energy function is defined to find an optimal contour C that minimizes $\varepsilon_x^{\text{Fit}}$ for all x in image I, which is expressed as (9), where $\nu|C|$ is an additional term to smooth the contour by penalizing its length. The global energy function can be minimized using standard gradient descent method [35]. The initial contours are rectangles centered at road seeds [see Fig. 1(d)]. LBF-Snake deforms the contours by minimizing the global energy and, finally, fits the contours on road boundaries [see Fig. 1(e)]

$$\varepsilon_x^{\text{Fit}}(C, f_1(x), f_2(x)) = \sum_{i=1}^{2} \lambda_i \int_{\Omega_i} K_\sigma(x-y)|I(y)-f_i(x)|^2 dy \tag{8}$$

$$\varepsilon(C, f_1(x), f_2(x)) = \int \varepsilon_x^{\text{Fit}}(C, f_1(x), f_2(x))dx + \nu|C|. \tag{9}$$

IV. Experiments

The proposed approach includes two parts: the patched DCNN model of RSEM and the heuristic algorithms of SVC, MMGF, and LBF-Snake. The performance and training sensitivity of the RSEM, and the parameter setting of the heuristic algorithms are the main considerations in our approach. In this section, we conduct comprehensive experiments to evaluate these issues.

A. Data Sets Overview

We evaluate our approach on two publicly available data sets, i.e., Google Earth [23] and Massachusetts road data set [2]. The Google Earth data set contains 224 very high resolution (VHR) images with a spatial resolution of 1.2 m which are manually labeled the accuracy road mask. The Massachusetts road data set is a well-known road extraction benchmark data set including 1172 images with resolution of 1500 × 1500, where 1108 images are for training, 14 images for validation, and 50 images for testing. In Massachusetts road data set, the road references generated from OpenStreetMap are suffered from missing and registration noise and the images are seriously covered by trees. The images in Google Earth are relatively better, but the roads are dense and the road color is variant dramatically between images, and even within an image. Fig. 7 shows some complex scenes in both data sets.

B. Evaluation Metrics

In this article, the classic metrics precision and recall are used to evaluate the performance of our approach on Google Earth data set, as shown in (10). While, on Massachusetts road data set, we computed relaxed precision and recall scores instead of exact ones since the road references are generated from road vectors which are only accurate up to a few pixels. Namely, the relaxed recall represents the fraction of true road pixels that are within 3 pixels of a predicted road pixel, while the relaxed precision represents the fraction of predicted road pixels that are within 3 pixels of a true road pixel. Relaxing the precision and recall measures is common practice when evaluating road detection systems [36]. The F1-score is the tradeoff of the precision and recall

$$P = \frac{\text{TP}}{\text{TP}+\text{FP}}, \quad R = \frac{\text{TP}}{\text{TP}+\text{FN}}, \quad F1 = \frac{2*P*R}{P+R}. \tag{10}$$

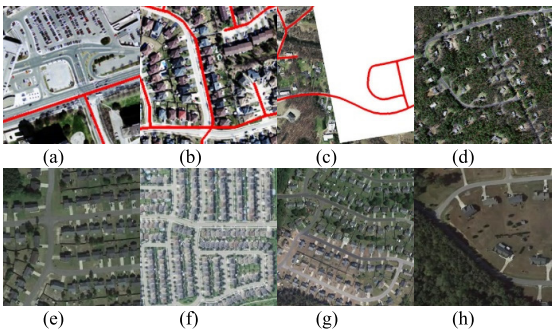


Fig. 7. Illustration of complex scenes in the test data sets, where (a)–(d) come from Massachusetts Roads Data set and (e)–(h) are sampled from Google Earth. (a) Example of omission noise. (b) Example of registration noise. (c) Example with large invalid area. (d) and (h) Examples of occlusion by trees. (e) and (f) show the intensity contrast. (g) presents examples of color variance in the same road segment.

C. Training Sensitivity of RSEM

Training a road center point estimation model is the key of our method. Only accurate road seeds and background points can provide good training samples for subsequent training of support vector classifiers. At first, in order to train the RSEM, the training samples are automatically calculated according the road mask. To generate training patches, the road mask is skeletonized; then, two patches are randomly cropped from the origin image and the skeletonized mask at the same position. If there are road segments in the patch, the skeleton inside the patch is used to build an undirected graph, and the center point of each connected component of the undirected graph is regarded as a road center point. Some samples are shown in the first row of Fig. 8. More samples are available *online*.[1] It is worth noting that the road center points must not be fixed at center of the patch, or the model will prefer to predict the patch center as the road center point in the prediction stage. If there is no road segment in the patch, an empty ground truth map is directly generated which is called a negative sample. The negative samples are used to train the model to output low road center confidence maps for background patches, which are very important for model to detect background points in our method.

As mentioned above, the road center points are accurate positions automatically calculated by computer. However, our method is more valuable in those scenes without road masks

[1]Online. [Available]: https://github.com/rob-lian/WeaklyRoadSeg/

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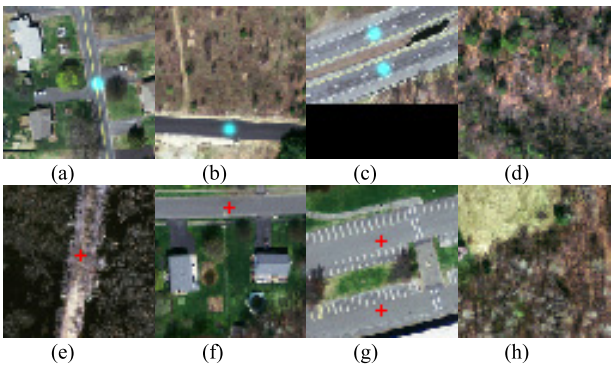


Fig. 8. Some training samples and visual prediction results of the road center estimation model applied to patches from the Massachusetts Roads data set. Some training samples shown in the composition of the patch and ground truth (first row). Some are the prediction samples, and the road centers are marked by red cross (second row).

where the training patches must be labeled manually. If the training of the DCNN model is sensitive to the accuracy of the road center annotations, which will dramatically decrease the value of our method, because we cannot request users to place road center points exactly on the road center. Thus, it is necessary to evaluate the sensitivity of this model to training samples. We hope that the training of the DCNN model is robust to the accuracy of the point annotations, so that users can choose the approximate locations of the road center points in a patch arbitrarily. To this end, during the automatic generation of training samples, the road center points are first vibrated on both sides of their precise positions along the road direction with the amplitude subjecting to the Normal distribution probability of 1/4 road length, and then they are shifted by the Normal probability of 1/2 road width along the perpendicular direction of the road. We generate three training data sets separately from Massachusetts road data set by setting σ of the Normal distribution to 1.0, 2.0, and 4.0. At the same time, in order to evaluate the real application, we implemented a road center point sampling software, and employed 78 students to manually sample a data set containing about 240000 training samples. Finally, we collected five training data sets including the precise road center point training data set. These training data sets are used to train road center estimation models separately. The loss curves of the training and validating are shown in Fig. 9, which indicate that all data sets can make the model converged, although the convergence is slower when more inaccurate samples are used. So, we conclude that the model is not sensitive to the accuracy of the road center annotations. Most importantly, we find that the model trained by manually sampled data set performs best. The reason is that the automatically sampled data set may contain some bad samples where the real road center points are wrongly calculated when the road skeleton in the patch is too complex, such as road junction, road loop, and so on. In addition, some roads are completely covered by trees, but the generated ground truths indicate that there are roads in the patches, which will make the model confused.

D. Kernel Function Selection in SVC

The basic kernels are: linear, polynomial, RBF, and sigmoid. In this article, we select RBF because it nonlinearly maps samples into a higher dimensional space, so it, unlike the linear kernel, can handle the case when the relation between class labels and attributes is nonlinear. Furthermore, the linear kernel is a special case of RBF with some parameters (C, γ). In addition, the sigmoid kernel behaves like RBF for certain parameters, and the polynomial kernel has more hyperparameters to be fine-tuned than the RBF kernel [37]. To justify the advantages of RBF, we compared the resulting segmentation using different kernel functions. As can be seen from Table II, on the Google Earth, SVC performs well with any kernel function of RBF, Polynomial, or Sigmoid except for Linear. SVC with linear kernel always introduces a lot of false positives (FPs) wrongly identifying the roofs as roads which results in low precision. On the Massachusetts road data set, SVC performs better using RBF than using any other kernel functions.

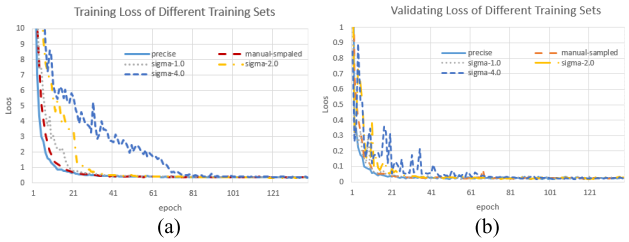


Fig. 9. Training and validating loss curves are shown in (a) and (b), respectively. The curves of precise, manual-sampled, sigma-1.0, sigma-2.0, and sigma-4.0 in (a) represent the training losses of the models trained by the precise points, manually sampled points and the points deviated subject to Normal distribution in which σ is set to 1.0, 2.0, and 4.0. The curves in (b) have the corresponding meanings of validation loss.

TABLE II
FINAL SEGMENTATION RESULTS WITH DIFFERENT KERNEL FUNCTIONS IN SVC

Kernel	Mass. Road set	Google Earth
RBF	P:84.5, R:82.2, F1:83.1	P:91.1, R:92.5, F1:91.8
Polynomial	P:75.4, R:80.6, F1:77.3	P:90.8, R:92.7, F1:91.7
Sigmoid	P:94.2, R:63.9, F1:75.6	P:90.5, R:92.6, F1:91.5
Linear	P:65.0, R:82.3, F1:71.4	P:73.9, R:85.3, F1:78.8

E. Parameters Study

There are a number of hyper parameters introduced in the pipeline of our approach, we list them as below: sliding step M in RSEM; α and C in SVC; λ, θ, ψ, σ, and γ in MMGF; ν and iteration times T in LBF-Snake. In order to evaluate the robustness of our approach with different hyper parameter settings and research the feasible parameter settings when our approach is used in brand new data sets, we conduct extensive experiments with different parameter settings on both of the test data sets. We evaluate all the parameters except for λ, ψ, σ because these three parameters are fixed, as described in Section III. Theoretically, RSEM produces more road seeds and proposals with smaller M, but consumes more time. MMGF is more robust to road direction with finer granularity of θ and more feasible to road intensity inhomogeneity with smaller γ. The fitted contours are smoother but prone to shrink from road edges with larger ν and T. α and C need to be grid searched for the optimal combination [38]. The experimental results are shown in Table III where the performances are the average F1-scores in percentage on the test sets. As can be

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TABLE III

ROBUSTNESS OF PARAMETERS EVALUATED ON GOOGLE EARTH AND MASSACHUSETTS ROAD DATA SETS. γ_1, γ_2, γ_3, AND γ_4 ARE [0.1], [0.1, 0.2], [0.1, 0.2, 0.4], AND [0.1, 0.2, 0.4, 1.0], RESPECTIVELY. Θ_{intv} MEANS THAT Θ IS A SET RANGING FROM 0 TO 180 WITH THE INTERVAL Θ_{intv}. NP_RD AND NP_BG REPRESENT THE AVERAGE NUMBER OF ROAD AND NONROAD POINTS DETECTED IN EACH IMAGE, RESPECTIVELY. ST AND FT MEAN THE AVERAGE SEARCH TIME OF RSEM AND THE AVERAGE FIT TIME OF LBF-SNAKE FOR EACH IMAGE IN SECOND. *X|Y* MEANS THAT THE VALUES IN THE BELOW CELLS ARE SET TO PARAMETER X, AND THOSE IN RIGHT CELLS ARE SET TO PARAMETER Y. ALL THE SCORES ARE F1-SCORES IN PERCENTAGE ACHIEVED BY OUR APPROACH WITH THE CORRESPONDING PARAMETER COMBINATION EXCEPT FOR THE COLUMNS OF ST, FT. THE "ADA" IN SVC MEANS α AND C ARE ADAPTIVE FOR EACH TEST IMAGE WHICH ARE GRID SEARCHED

	Google earth dataset					Massachusetts road dataset				
	α=0.1, C=10, γ= γ_4, θ_{intv}=10, λ=30, σ=2, ψ=0, ν=650, T=50					α=0.333, C=1.0, γ= γ_1, θ_{intv}=5, λ=30 σ=2, ψ=0, ν=325, T=50				
RSEM	*M*	*NP_rd*	*NP_bg*	*ST(s)*	*F1*	*M*	*NP_rd*	*NP_bg*	*ST(s)*	*F1*
	16	982.7	1440.7	16.11	90.25	16	2692.4	5407.6	357.8	82.40
	32	248.0	369.2	4.16	91.77	32	683.7	1341.3	89.2	83.08
	48	110.9	171.0	1.90	89.37	48	303.8	596.2	37.5	74.71
	64	63.2	96.7	1.08	87.73	64	81.0	448.0	21.9	65.05
	M=32, γ=γ_4, θ_{intv}=10, λ=30, σ=2, ψ=0, ν=650, T=25					*M=32, γ=γ_1, θ_{intv}=5, λ=30, σ=2, ψ=0 ν=325, T=25*				
SVC	α | C	1	10	100	Ada	α | C	1	10	100	Ada
	0.001	89.35	89.64	89.17	-	0.001	81.36	81.53	81.44	-
	0.01	90.76	89.49	90.15	-	0.01	81.68	81.86	81.71	-
	0.1	91.15	91.62	91.57	-	0.1	81.94	82.10	82.32	-
	0.333	90.93	91.00	91.01	-	0.333	82.17	82.26	82.24	-
	Ada	-	-	-	91.26	Ada	-	-	-	82.12
	M=32, α=0.333, C=1.0, λ=30, σ=2, ψ=0, ν=650, T=25					*M=32, α=0.333, C=1.0, λ=30, σ=2 ψ=0, ν=325, T=25*				
MMGF	γ | θ_{intv}	*5*	*10*	*30*	*45*	γ | θ_{intv}	*5*	*10*	*30*	*45*
	γ_1	87.97	87.95	85.70	79.83	γ_1	82.17	82.22	78.63	73.69
	γ_2	90.30	90.42	90.50	89.87	γ_2	81.22	81.12	80.15	79.55
	γ_3	90.89	90.83	90.69	90.50	γ_3	79.93	79.77	78.92	78.40
	γ_4	91.30	91.34	90.66	90.33	γ_4	77.25	77.00	74.68	74.64
	M=32, α=0.333, C=1.0, γ= γ_4, θ_{intv}=10, λ=30, σ=2, ψ=0					*M=32, α=0.333, C=1.0, γ= γ_1 θ_{intv}=5, λ=30, σ=2, ψ=0*				
LBF-Snake	T|ν	325	650	975	*FT(s)*	T|ν	325	650	975	*FT(s)*
	10	90.57	90.20	89.97	0.78	10	80.68	79.28	78.61	2.86
	25	90.93	90.84	90.68	1.91	25	82.17	81.19	80.58	6.96
	50	90.84	90.89	90.95	3.64	50	82.63	82.05	81.45	14.13
	100	91.45	91.50	91.44	7.29	100	82.56	82.41	82.30	28.19

seen from Table III, M is the most important parameter in our approach, which determines the number of road seeds and road proposals and further determines the output of SVC and MMGF. The importance of M is more prominent in complex scenes. For example, when the value of M changes from 32 to 64, the F1-score on Massachusetts road data set decreases 18.03% comparing with 4.01% on Google Earth data set. However, smaller M means more computation requirement and our performances become worse when M is smaller than some threshold because SVC may fail to obtain the optimal hyperplane between the pixels of road and background when there are too many contradictory training samples. Our approach presents high robustness in the setting of α and C, and it is worth noting that the adaptive α and C for each test image which are selected by grid-search do not perform best, which further verifies the robustness of our approach. In MMGF, we found that our approach presents sensitive to the granularity of θ when γ is small, and γ should be set to a small value when the road intensity is inhomogeneous. Finer granularity of θ means more kernels in Gabor Filter; however, the convolution of MMGF can be accelerated by GPU, and we can set θ to a fine granularity with negligible increase of time requirement, such as ranging from 0° to 180° with 5° interval. The LBF-Snake can reach convergence in about 25 iterations, and the setting of the parameter ν affects hardly on our approach.

In summary, in most situations, M can be set to 32. α and C can be fixed as default values, i.e., *1/n* and 1.0 where n is the feature dimension of the training data, while ν can be set from 325 to 650, and T from 25 to 50. In MMGF, θ_{intv} can be fixed to a small value, such as 5 or 10. As for γ, when the road intensity is homogeneous, γ can be a set with more and larger values, and vice versa.

F. Road Segmentation Experiments

The first experiment is evaluated on Google Earth data set. In this experiment, the high and low confidence thresholds for road seeds and background points prediction are 0.3 and 0.2, respectively. The kernel function of SVC is RBF with the penalty coefficient of $C = 10.0$, γ of MMGF is set to [0.1, 0.2, 0.4, 1.0], and ν and T are set to 650 and 50 in LBF-Snake. The RSEM is trained by the auto-generated training data set in which 100 patches are randomly sampled from each image, and the point annotations are vibrated subjecting to a Normal distribution with $\sigma = 4.0$. The patch size is 64×64. The model is trained 20 epochs with the mini-batch size of 64 and the learning rate 1e-5. In order to verify the accuracy of RSEM, we set the confidence threshold to 0.25 and extract local peaks from the confidence map predicted by RSEM. If the peak values are greater than the threshold, the corresponding points are regarded as road seeds. Contrarily, there is no road in the patch if all the peak values are below the threshold. The model achieves F1-score 95.14% on the test data set. The F1-score is calculated by (10), where P, R, TP, FP, and FN represent precision, recall, true positive, false positive, and false negative, respectively. Here, TP means that all the road seeds inferred by RSEM fall on the road, or RSEM correctly judges a background patch. FP indicates that any of the predicted road seeds outside the road or wrongly output road seeds when no road exists in the patch. FN represents that the model mistakenly regards a patch as background when the patch contains roads. The processing steps of an image of Google Earth with resolution 1039×678 are shown in Fig. 10(a)–(g). As can be seen that the roads are dense, and the colors of some house roofs and roads are similar. First, the road and nonroad seeds are searched according to the algorithm described in Section III-B, as is shown in Fig. 10(a). Second, using these seeds, the trained SVC classifies all pixels into road and nonroad, which are represented by red and white in Fig. 10(b). The road proposals are also regarded as road pixels. Based on the pixel-wise classification, MMGF is used to assess the road potential, as shown in Fig. 10(c), it can be seen that the road area has higher road probability. Third, rectangular initial contours centered at each road seeds are generated [see Fig. 10(d)]. Fourth, the final contours are

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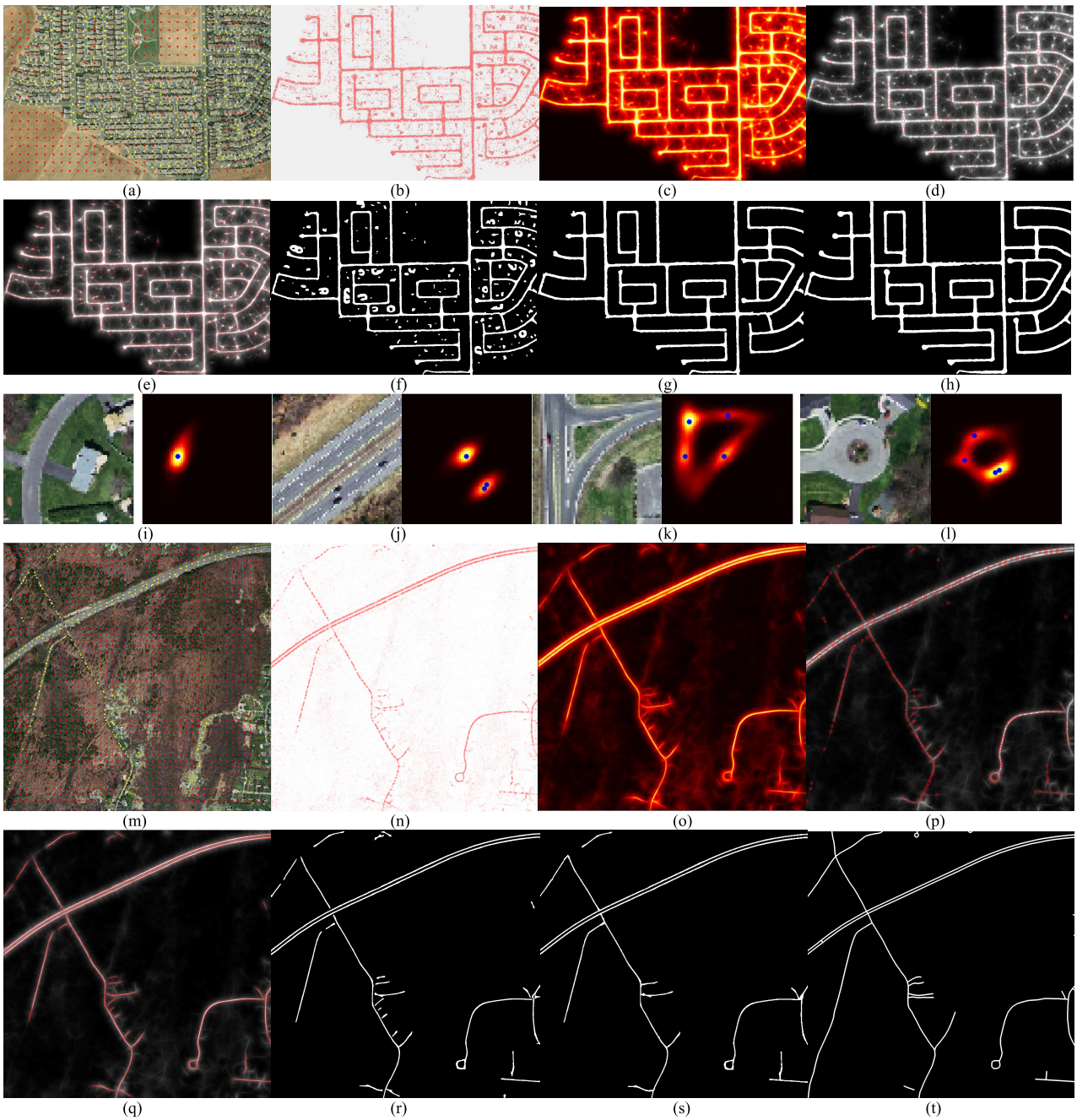


Fig. 10. Two experiments performed in the Google Earth data set (a)–(h) and Massachusetts road data set (m)–(t). (a) and (m) are two images marked by roads seeds (yellow) and background points (red), and the image sizes are 1039 × 768 and 1500 × 1500, respectively. (b) and (n) are the pixel-wise classifications produced by SVC combining the road proposals estimated by RSEM. (c) and (o) are the road potentials generated by MMGF. (d) and (p) illustrate the initial contours around the road seeds. (e) and (q) are the final contours fitted by LBF-Snake. (f) and (r) are the road regions created by the final contours. (g) and (s) are the road regions refined by post-processing. (h) and (t) are the ground truths. (i)–(l) are some examples of the output of RSEM where the hot areas are the road proposals and the blue dots are road seeds, the left image is the input patches, and the right map is the corresponding output in each pair.

obtained by executing LBF-Snake for 50 iterations, as shown in Fig. 10(e), and the segmentation is achieved according to the fitted contours, as shown in Fig. 10(f). The small fake road regions can be easily removed by a simple post-processing as follows: remove the road regions without road seeds and those with small area, then dilation. As shown in Fig. 10(g), Fig. 10(h) shows the ground truth, which indicates that the extracted road segmentation is very close to the ground truth.

The second experiment is performed on Massachusetts road data set. Since the images are much more complicated and the colors of road and nonroad are worse linear Separable than Google Earth, SVC produces much more wrong classifications. Thus, the road proposals predicted by RSEM become especially important in this experiment. Fig. 10(i)–(l) shows four examples of the output of RSEM, where the blue dots are the road seeds and the hot areas represent road proposals. Even so, the preliminary classification is still poor and the distribution of road points is sparse, as shown in Fig. 10(m) and (n). Therefore, γ of MMGF should be set to a small value, e.g., 0.1, such that the road probability of

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TABLE IV
ANNOTATION TIME COMPARISON BETWEEN FULL SUPERVISION AND POINT SUPERVISION

Dataset	Fully supervision	Point supervision
Google Earth	852.0 sec/img	62.2 sec/img
Massachusetts	645.2 sec/img	65.1 sec/img

each point can be verified by farther collinear road points; otherwise, many blocks with high road probability will be generated by MMGF, which will further make the contours fitted by LBF-Snake discontinuous. However, too small γ will cause the road probability low at the T-junctions which will make the final road contours cracked at these places, as shown in Fig. 10(q), we solved this problem in post-processing as below: removing the regions without road seeds, and then connecting the small gaps. The RSEM is trained by the manually sampled training patches with size of 64 $\times$ 64 and achieved 92.1% F1-score when the confidence threshold is set to 0.12. During the search of road and background points, the high and low confidence thresholds are set to 0.1 and 0.06, respectively. Fig. 10(m)–(s) shows the processing steps of an image with resolution 1500 $\times$ 1500 in Massachusetts road data set, and Fig. 10(t) shows the road references. As can be seen from Fig. 10, although the result is quite good, there are still some roads that cannot be extracted due to the inhomogeneity of road color and the severe tree coverage.

V. COMPARISONS

Comparing Algorithms: We compare our method with three state-of-the-art road extraction methods based on the fully supervision. The comparing algorithms are listed as follows.

1) CasNet: A road extraction method proposed by Cheng *et al.* [23], in which two UNet-like networks are cascaded for road segmentation and road centerlines extraction. We take the output of the first network as the road segmentation results.
2) Ventura: A road centerline tracking method proposed by Ventura *et al.* [39], in which a patched DCNN model is designed to infer the border points that connect the patch center by roads. The global network is tracked by iterative prediction of the local connectivity guided by a coarse road map produced by another dense semantical segmentation network. We dilate the extracted road centerlines to obtain the road regions.
3) ASPP-UNet: A road extraction method based on a modified UNet proposed by He *et al.* [40], in which the bridge part is replaced by an atrous spatial pyramid pooling (ASPP) module.
4) Ours: As Fig. 2 shows, RSEM, SVC, MMGF, and LBF-Snake are employed to obtain the final road regions. The parameters of our method are the same for each image in the same data set, as described in Section III.

A. Annotation Cost Comparison

To fairly evaluate our performance, the proposed method is compared with three state-of-the-art fully supervised methods from two perspectives: unlimited budget and fixed budget. The budget means the labor time for labeling training data. Since the authors of the test data sets did not report the time for the road labeling, we manually relabeled some of them to estimate the average time for labeling road mask of an image. For labeling the pixel-wise road mask in Google Earth, a universal tool [41] for pixel-wise labeling is used to manually annotate roads. As for Massachusetts road data set where the road masks were dilated from road center lines, R2V [42] is employed to extract road center lines in interactive mode. This allows us to properly compare this supervision setting with human points. For sampling the training patches of our RSEM, as mentioned in Section IV-C, a point sampling tool [43] is developed to manually place the road center points and background points. we organize 78 students, assign 2 randomly selected images to each student on each test data set, and count the time spent on the point sampling on both data sets, pixel-wise labeling on Google Earth data set and road center lines extraction on Massachusetts road data set. Finally, the respective average times are regarded as the final labor cost. Note that, the patch-point training pairs used in Ventura [39] are autogenerated from road masks, and the annotation times of Ventura's method are calculated as the other fully supervised methods. As can be seen from Table IV, the pixel-wise annotations consumed more time, although the image sizes in Google Earth are much smaller. For point annotation, the number of the points sampled from each image is fixed 50, so no matter which data set, the label time is similar. Although the image sizes of the Massachusetts road data set are much larger and the roads are more complex, it takes less time because we just extract the road centerlines instead of marking the road mask. We enforce quality control to the annotations, i.e., less than 5% of the annotated pixels are out of the roads in Google Earth, extract all the road center lines of an image in the Massachusetts data set, and all the sampled road points should be located on the road area in our method. As can be seen from Table IV, Point supervision is more than 10$\times$ cheaper than full supervision.

B. Comparison on Unlimited Budget

First, we compare the performances of our model with three state-of-the-art methods without considering the annotation budget, that is, all the training data are used to train the comparing models. This comparison shows the performance gaps between our weakly supervised model and the fully supervised ones.

Fig. 11 shows the qualitative comparisons between ours and three state-of-the-art algorithms. CasNet performs poor in complex scenes, such as Massachusetts road data set, because it is built directly on the standard UNet which is likely to miss unobvious targets. ASPP-UNet improved from the standard UNet, especially the replacement of the middle part with ASPP unit, which can effectively extract small objects in the image and make the road network more complete. Ventura performs relatively poor because it predicts the connectivity between boundary points and the center point of a patch. In complex scenes, some boundary points are wrongly predicted, which introduces a number of FP, especially in the areas of broad crossroads and parallel roads. Conversely, once some boundary points fail to be inferred, many subsequent patches will be neglected in the tracking process according

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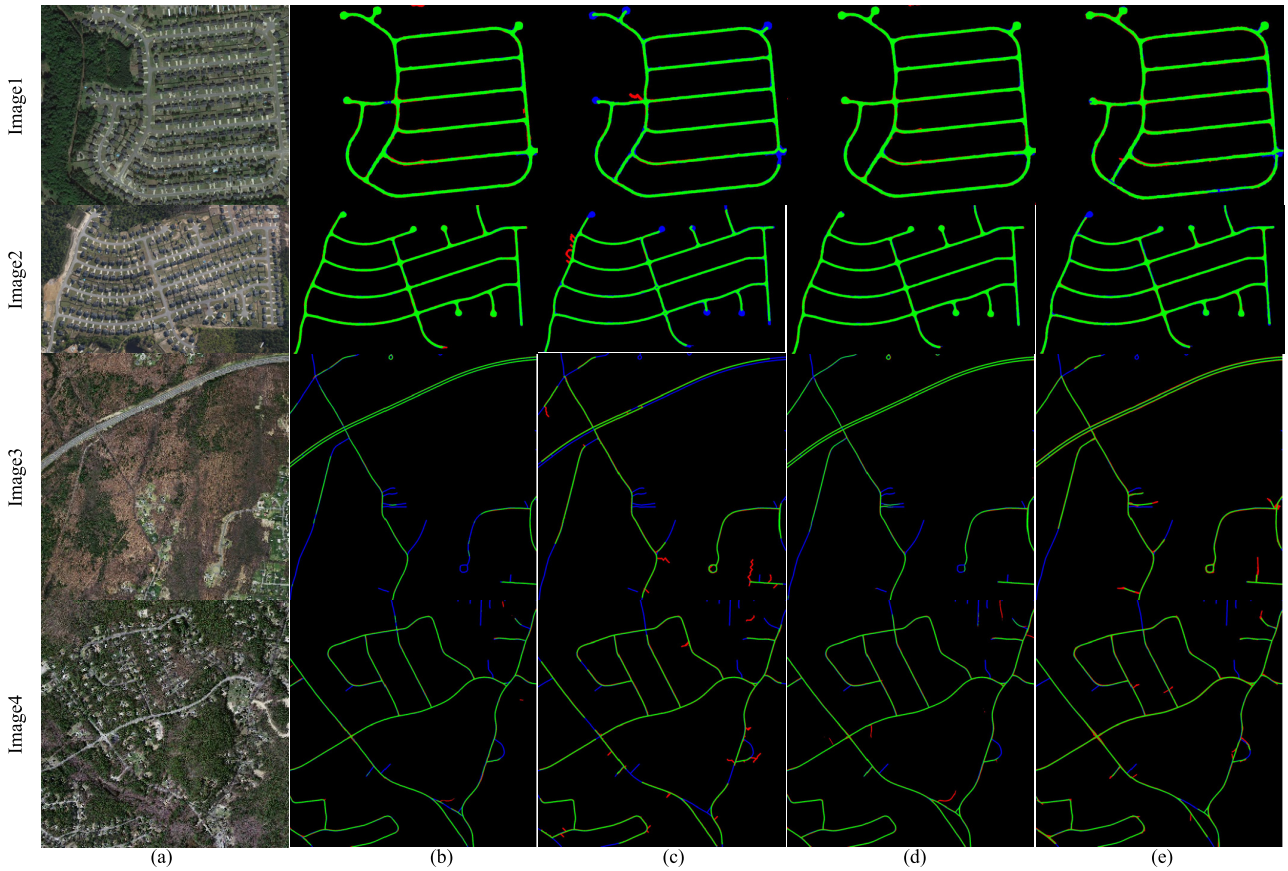


Fig. 11. Qualitative comparisons of road extraction results with different comparing algorithms. The first two rows are from Google Earth, the last two rows are from Massachusetts road data set. (Green) TP. (Red) FP. (Blue) FN.

to its tracking strategy. Although our segmentation results are not the best, our method extracts road regions quite well, that is, in Google Earth data set, the integral road network is segmented associated with some slight spurs and some negligible FPs and FNs along the roadsides, and in Massachusetts road data set, a number of FNs are introduced in our results due to the inhomogeneous road color and the shade of trees, which also defects the other comparing methods.

Table V shows the quantitative comparisons. The images are those shown in Fig. 11. As can be seen from Table V, performance of all methods is better on Google Earth data set, because the road color is more homogenous and the roads are less covered by trees compared with Massachusetts road data set. Most importantly, the ground truths in Massachusetts road data set are not accurate, some of them are missing, and some are ill-registered, which further decreases the performance of all methods and especially worse to our method, because the intensity of impervious surface is the most important evidence for road detecting in our scheme. Checking the results of our approach in Fig. 12, we find that roads really exist in the regions of FP (red) and no road exists in the FN areas (blue). Thus, higher scores do not necessarily mean better segmentation in Massachusetts road data set. From the qualitative and quantitative comparison, we can come to a conclusion that our method achieves pretty good results. The most important is that our results are close to the fully supervised methods with less than one-tenth labor cost.

TABLE V

QUANTITATIVE COMPARISONS AMONG DIFFERENT METHODS ON ROAD SEGMENTATION WITHOUT CONSIDERING LABEL BUDGET, WHERE THE VALUES IN BOLD ARE THE BEST AND THE UNDERLINED VALUES ARE THE SECOND BEST

DS	Image	Method	P	R	F1	Supervision
Google Earth	Image1	CasNet	90.8	96.4	93.5	Full
		Ventura	**93.6**	86.9	90.1	Full
		ASPP-UNet	92.1	**96.9**	**94.4**	Full
		Ours	89.3	93.4	91.3	Point
	Image2	CasNet	92.6	**97.1**	94.8	Full
		Ventura	94.8	84.3	89.2	Full
		ASPP-UNet	**95.5**	95.1	**95.3**	Full
		Ours	93.4	91.0	92.2	Point
	Avg. (test set)	CasNet	91.6	94.8	93.2	Full
		Ventura	86.2	84.5	85.3	Full
		ASPP-UNet	**96.3**	**95.9**	**96.1**	Full
		Ours	91.1	92.5	91.8	Point
Massachusetts	Image3	CasNet	**100**	75.2	85.8	Full
		Ventura	92.2	62.5	74.5	Full
		ASPP-UNet	99.9	85.2	**92.0**	Full
		Ours(acc)	96.5	**86.6**	91.3	Point
	Image4	CasNet	**98.6**	91.3	94.8	Full
		Ventura	94.4	80.9	87.1	Full
		ASPP-UNet	96.6	**93.5**	**95.0**	Full
		Ours(acc)	96.6	90.1	93.2	Point
	Avg. (test set)	CasNet	**97.5**	81.5	88.4	Full
		Ventura	87.8	72.4	78.6	Full
		ASPP-UNet	94.9	**88.6**	**91.4**	Full
		Ours(acc)	84.5	82.2	83.1	Point

C. Comparison on a Fixed Budget

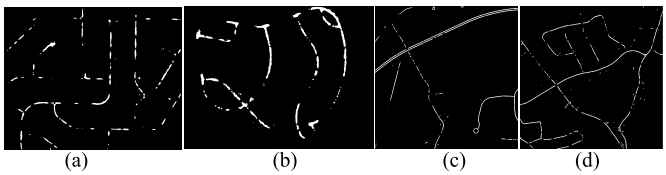
Given a fixed annotation time budget, what is the right strategy to obtain the best road segmentation model possible?

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(a) (b)

Fig. 12. (a) Image superimposed by origin image, segmentation, and ground truth, where green, red, and blue represent TPs, FPs, and FNs, respectively. (b) Close-ups where we find that roads (mainly the aisles around the houses) really exist in the FP regions and no road in some FN regions (usually caused by the noise of road references).



(a) (b) (c) (d)

Fig. 13. Some failed examples predicted by ASPP-UNET trained by limited training data. (a) and (b) are the results on Google Earth data set. (c) and (d) are the samples on Massachusetts road data set.

TABLE VI

QUANTITATIVE COMPARISONS AMONG DIFFERENT METHODS ON ROAD SEGMENTATION GIVEN A FIXED BUDGET, WHERE THE VALUES IN BOLD ARE THE BEST AND THE UNDERLINED VALUES ARE THE SECOND BEST

Dataset	Supervision	Method	Precision	Recall	F1-score
Google Earth dataset	Full (15 imgs)	CasNet	**92.6**	65.5	71.2
		Ventura	49.4	56.3	52.0
		ASPP-UNet	90.5	61.3	70.0
	Point (all)	Ours	91.1	**92.5**	**91.8**
Mass. Road dataset	Full (112 imgs)	CasNet	**93.2**	67.5	77.7
		Ventura	64.4	69.7	66.5
		ASPP-UNet	88.4	73.9	80.0
	Point (all)	Ous	84.5	**82.2**	**83.1**

We investigate the problem by fixing the total annotation time. For each fully supervised method, we compute the number of images N that it is possible to be labeled in that amount of time, randomly select N images from the training set, use them to train segmentation models, and measure the performances on the test set, as done in [7]. On Google Earth dataset, the time for sampling all the training patches for our model can be used to annotate the pixel-wise road masks for about 15 images, and the label cost for our model on Massachusetts road dataset is enough to extract road centerlines for about 112 images. On such small data sets, we do the data augmentation including random cutting, flip, image scaling, and shifting to mitigate the problem of overfitting. Table VI reports both the number of images N and the resulting performances. All the fully supervised models are inevitably over-fitted. CasNet and ASPP-UNET tend to relay more on road features which result in high precision but low recall. Ventura detects a lot of FP due to the massive wrong prediction of border points which leads to very low precision. The overfitting also can be found in Fig. 13, where many road regions are missed in the results. All the quantitative and qualitative comparisons show that our method outperforms the fully supervised methods given a fixed budget. Our method provides a better tradeoff between annotation time and segmentation results.

VI. CONCLUSION

A weakly supervised road segmentation method using point annotations is proposed in this article, which combines the powerful feature representation of DCNN and the interpretability of heuristic algorithms. Our method introduces point-level training data into road segmentation in remote sensing images for the first time and achieved good results. This work indicates a new research methodology for road segmentation that roads can be identified by pointing out where are the roads instead of marking all the roads, just like people teach babies. Our method also has shortcomings. For example, the processing time is longer than semantic segmentation DCNN model, and some parameters of our method need to be slightly adjusted according to the characteristics of the test data set.

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